Part 2 | Supervised Learning Baseline

Naive Bayes vs. decision tree: predicting student dropout and academic success

CS-13410 Introduction to Machine Learning · Fall 2026 · Assignment 2 | **Instructor:** Salman Ali
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Repository: github.com/maangatmaangat7-max/IML-7B-Project (notebook: *assignment 2/Part 2 - Supervised Learning Baseline.ipynb*)

Setup

Same data and split as Part 1: UCI dropout dataset, 4,424 students, target Dropout / Enrolled / Graduate, stratified 80/20 split (3,539 train, 885 test, seed 42). Primary metric is macro F1. Preprocessing sits *inside* each model pipeline, so every cross-validation fold fits its own imputers, clip fences, scalers and encoders. Models: Gaussian NB on the full Part 1 matrix, Gaussian NB on the 21 numeric columns, Bernoulli NB on quantile-binned and one-hot features, an unpruned decision tree, and a decision tree tuned (criterion, depth, leaf size, class weight) with **nested** cross-validation, plus a majority-class reference.

Hand-worked split (entropy and Gini)

Root split chosen on the training data: **2nd-semester approved units ≤ 3.5**. Class counts (Dropout / Enrolled / Graduate): left 844 / 235 / 96 (n = 1,175); right 293 / 400 / 1,671 (n = 2,364); parent 1,137 / 635 / 1,767 (n = 3,539).
Entropy: H(parent) = 1.4713 bits; H(left) = 1.1025; H(right) = 1.1608; weighted children = (1175/3539)(1.1025) + (2364/3539)(1.1608) = 1.1415; **IG = 1.4713 − 1.1415 = 0.3299 bits**.
Gini: G(parent) = 0.6153; G(left) = 0.4374; G(right) = 0.4564; weighted = 0.4501; **ΔG = 0.1652**. The arithmetic matches scikit-learn's impurity values. A Gini stump picks the same feature at ≤ 4.5 (ΔG = 0.1660), so the criteria agree on the feature but not the threshold.

Cross-validation and test results

5-fold stratified CV repeated 3 times on the training split (identical folds for every model). The unpruned tree was not scored on the test set. The test set was scored once, after all choices were fixed.

Model	CV macro F1 (15 folds)	CV ROC-AUC	CV recall Enrolled	Test macro F1	Test ROC-AUC
Majority class (reference)	0.222 ± 0.000	0.500	0.000	–	–
Gaussian NB (all features)	0.217 ± 0.010	0.704	0.933	0.219	0.707
Gaussian NB (numeric only)	0.651 ± 0.012	0.827	0.454	0.660	0.843
Bernoulli NB (binned)	0.671 ± 0.013	0.858	0.579	0.678	0.860
Decision tree (unpruned)	0.626 ± 0.013	0.734	0.379	–	–
Decision tree (tuned, nested CV)	0.676 ± 0.015	0.847	0.549	0.674	0.848

Which model is better, and why

The tuned tree and Bernoulli NB are statistically tied (macro F1 difference 0.005, corrected paired p = 0.69; ROC-AUC p = 0.15), and test scores agree (0.674 vs 0.678). We do not claim either is better. Both clearly beat the unpruned tree ($p < 0.001$), which memorises the training data: training macro F1 reaches about 0.96 at depth 16 while validation F1 peaks near depth 6. Tuning (entropy, depth 8, ≥ 30 samples per leaf, balanced class weights) fixes this.

Naive Bayes assumes conditionally independent features, which is violated here (semester-1 and semester-2 counts, parents' qualifications), yet it ranks students well (ROC-AUC 0.83-0.86) because only the ordering of posteriors matters. **Representation mattered more than the NB variant:** Gaussian NB on the full one-hot matrix fails (below the majority-class reference), because 34% of its within-class variances are under 10^{-3} and a normal density that narrow turns any rare category into overwhelming evidence; on numeric columns only it reaches 0.651, and Bernoulli NB on binned features reaches 0.671.

All models are weakest on *Enrolled*: the tuned tree's test recall is 0.65 but precision only 0.38. The tree's importances are dominated by the Part 1 feature 2nd-semester approval rate (about 0.66), then tuition fees up to date. Both models use semester-2 information, so they are end-of-year-1 early-warning tools, not enrollment-time predictors. We would choose the tree for explainability and Bernoulli NB for speed (0.1 s vs 22 s per fit including tuning). *Caveat:* with 159 Enrolled test students, differences of ±0.01 macro F1 are within noise. Part 3 (ensembles, tuning) tests whether the 0.68 ceiling is the data or these simple baselines.